

Course: B. Tech.

Branch: All Branches

Semester: II

Subject Code & Name: BTES205E Energy and Environment Engineering

Max Marks: 60

Date: 24/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain the working of Gas turbine-based power plant with a neat sketch.	CO1	6
B) Explain Hydro-power generation with schematic arrangement. Write down its advantages.	CO1	6
C) Explain with neat labelled diagram: Steam power plant. Write down environmental aspects considered for selecting the sites and locations for Thermal power plant.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Compare the relative characteristics of HAWT and VAWT in wind power plant.	CO2	6
B) What is Bio-mass? Write construction and working of bio-gas plant, with a neat diagram. Also write down the advantages of it.	CO2	6
C) What is OTEC? Explain closed cycle OTEC. Write down its advantages.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Write down the difference between energy conservation and energy efficiency with a suitable example.	CO3	6
B) Explain how the energy conservation can be practiced in the following: (any two of these)	CO3	6
a. Ventilation		
b. Air conditioning		
c. Boilers		
C) Define energy pyramid. Explain the measures to be taken to reduce the energy consumption in refrigerator.	CO3	6
Q.4 Solve Any Two of the following.		12

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| A) What is the difference between primary pollutants and secondary pollutants?
Give the examples of each. List out the effect caused by Polluted air. | CO4 | 6 |
| B) Explain shortly: a. Green House gases b. Eutrophication | CO4 | 6 |
| C) List out the various causes of noise pollution. Write down the various effects of noise pollution on Human health. | CO4 | 6 |

Q. 5 Solve Any Two of the following. **12**

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| A) Explain the various causes and effects of water pollution. | CO4 | 6 |
| B) List the types of solid waste based on its sources. Explain the different methods of solid waste disposal. | CO4 | 6 |
| C) What do you mean by radiation pollution. Write down the effect of radiation pollution. | CO4 | 6 |

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